

Synthesis, Student Research Designs, and the Bridge to Labor II

Labor I – Week 13

Teaching deck draft

Why Week 13 is a synthesis and research-design week

- Labor I has built the worker side of labor economics: labor supply, human capital, job value, households, inequality, discrimination, mobility, policy, and behavioral wedges.
- Week 13 converts topic mastery into research capability.
- The capstone question is: what is the object, what is the mechanism, what design isolates it, and when do firms or equilibrium responses become central?

The full Labor I architecture

- Week 1: measurement and labor-market objects.
- Weeks 2–3: static and dynamic worker choice.
- Weeks 4–5: skill accumulation, wages, and returns.
- Weeks 6–7: households, care, amenities, and total job value.
- Weeks 8–10: inequality, discrimination, sorting, and mobility.
- Weeks 11–12: policy design, behavioral wedges, and interpretation.

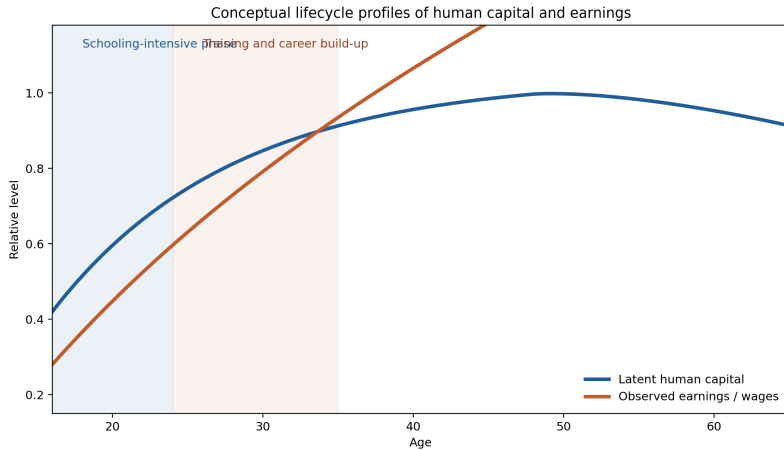
Unified worker-side framework

$$V_{it}(x_{it}) = \max_{h_t, s_t, j_t, m_t} \{u(c_t, 1 - h_t, a_{j_t}, \ell_t; \theta_i) - \chi_i(h_t, s_t, m_t) + \beta \mathbb{E}_t[V_{i,t+1}(x_{i,t+1})]\}$$

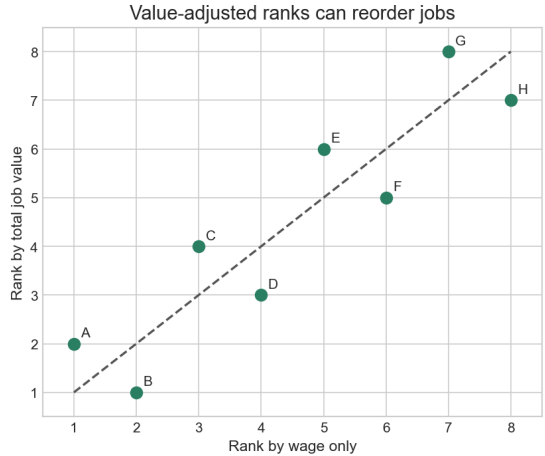
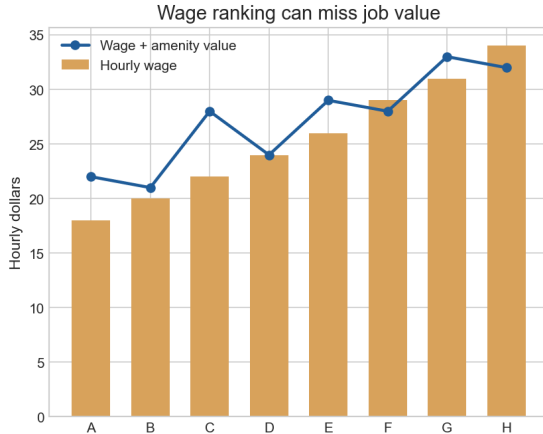
$$k_{i,t+1} = g(k_{it}, s_t, x_{it}, \varepsilon_{i,t+1}) \quad J_{ij} = w_{ij} + A_{ij} - C_{ij}$$

- Hours, skill investment, job choice, and mobility are the recurring Labor I margins.
- Policy, household structure, information, and behavioral wedges shift the feasible set or the perceived set.

Human capital is the dynamic backbone



Wages are not the whole object



What makes a good labor research question?

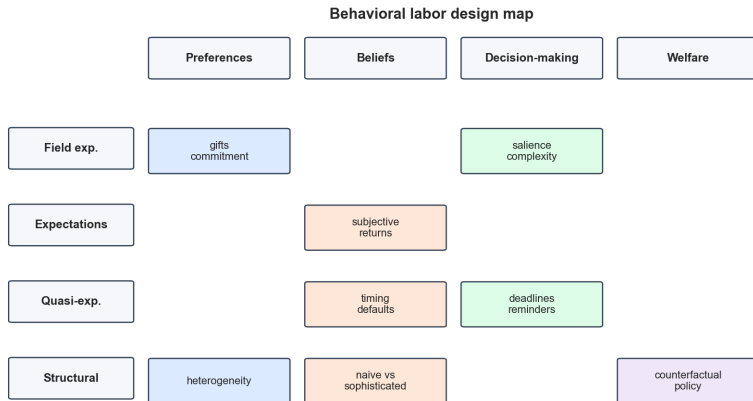
- A topic is not a question.
- A descriptive pattern is not a mechanism.
- A mechanism is not an identification strategy.
- A good labor question names one object, one mechanism, one population, and one contribution.
- A good Week 13 question also says whether it is partial equilibrium or already needs Labor II tools.

Mechanism–data–design–contribution pipeline

$$\tau = \mathbb{E}[Y_i(1) - Y_i(0)] \quad P(j \mid i) = f(k_i, z_i, a_j, w_j, d_{ij})$$

- Object: what outcome or estimand do we care about?
- Mechanism: what margin is supposed to move?
- Data: which unit and which measurement technology fit the object?
- Design: what variation or model discipline isolates the mechanism?
- Contribution: what would labor economists understand differently if the result were true?

The design problem is a matching problem



No single design is enough when the mechanism could be preferences, beliefs, or perception.

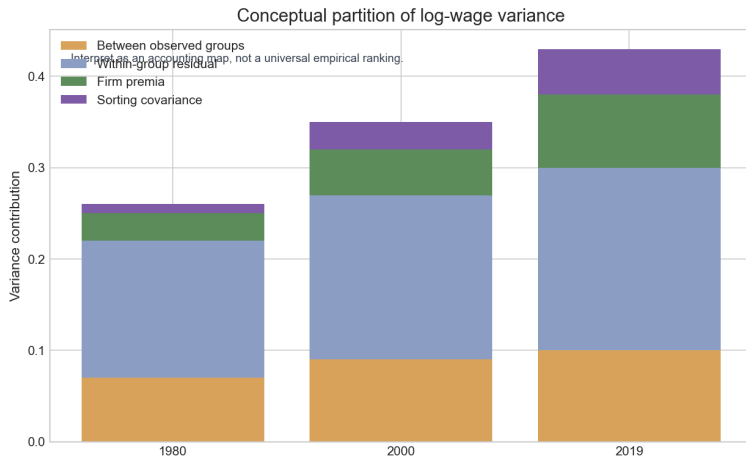
Example research templates

- Labor supply under schedule volatility, benefit design, or information.
- Human-capital investment under beliefs, liquidity, or timing frictions.
- Wage determination with selection, sorting, or firm premia.
- Household and care responses to policy or shocks.
- Job amenities, flexibility, and total job value.
- Inequality, discrimination, mobility, and behavioral response.

Common Week 13 failure modes

- Topic without a question.
- Mechanism without design.
- Design without labor contribution.
- Too many outcomes and no primary estimand.
- Hidden equilibrium issue.
- Literature survey posing as an idea.

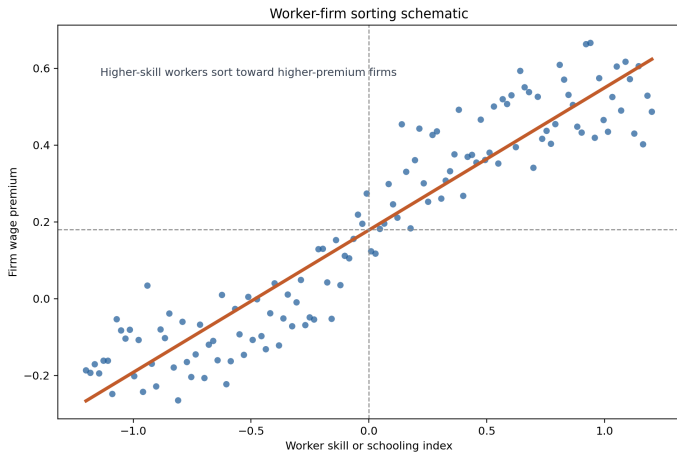
Inequality is an accumulation and assignment outcome



$$w_{ij} = \phi_i + \psi_j + \varepsilon_{ij} \quad \text{or} \quad M(U, V)$$

- Labor I asks how workers choose, invest, sort, and respond.
- Labor II asks what changes once firms choose wages, vacancies, schedules, training, and organizational rules.
- Search frictions, monopsony, personnel economics, institutions, and aggregate adjustment all live on this bridge.

Where Labor I becomes Labor II



Student memo template

- One sentence: “This paper asks...”
- One paragraph on the object and the population.
- One paragraph on the mechanism and the main alternative.
- One paragraph on data and measurement.
- One paragraph on design or model discipline.
- One paragraph on the labor contribution.
- One paragraph on whether the idea remains Labor I or already requires Labor II.

Where Labor I fits inside modern labor economics

- Labor I is the worker-side core of the field.
- Modern labor economics studies worker choices jointly with firms, institutions, and shocks.
- The same worker-side question can become a firm-side, search, bargaining, or equilibrium question once adjustment margins are opened.
- Week 13 should leave students with closure on the first semester and a map for where the second semester begins.

Takeaways

- Labor I is coherent once we organize it around worker-side objects and margins.
- Strong labor ideas move from topic to mechanism, data, design, and contribution.
- Many strong Labor I projects are partial-equilibrium by design, but they should name where Labor II begins.
- The capstone deliverable is a feasible research memo, not a giant agenda.

Appendix: anchor-paper menu

- Card (1999): returns and identification.
- Autor, Katz, and Kearney (2008): inequality architecture.
- Card et al. (2018) and Song et al. (2019): worker–firm bridge.
- Neumark (2018): discrimination and design discipline.
- DellaVigna (2009, 2018): behavioral wedge and research design.